Roll No: | | | | | | | | |

MID SEMESTER EXAMINATION Sept-2025

Name of The Program : M.Tech (Structural. Engg)
Name OF the Course : Design of Bridges
Time: 90 Min

Semester : III
Course Code : MSE 328
Max Marks : 50

Note: i) Answer **all the questions** by **choosing any one of the sub questions.**
ii) Each question carries 10 Marks.
iii) IS code 456-2000, IRC 5,6,12,112 supplied by the department is permitted for design use.

Q1 **Answer any one.** **(1X10=10 Marks)**

(a) Discuss the role of characterizing parameters for longitudinal bending moments in highway bridges, particularly in multi-cell box girder decks? *(CO1)*

(OR)

(b) Explain Courbon's method of analysis for RC beam and slab bridge decks, including its basic assumptions and limitations? *(CO3)*

Q2 **Answer any one.** **(1X10=10 Marks)**

(a) A bridge deck is subjected to transverse shear and bending moments. Interpret the combined effect of these responses on load distribution and deck cracking? *(CO1)*

(OR)

(b) Analyze the role of reaction factors for longitudinal girders in Courbon's method and evaluate how they influence load distribution among girders? *(CO3)*

Q3 **Answer any one.** **(1X10=10 Marks)**

(a) Evaluate the significance of longitudinal twisting moments in skew bridges and their impact on structural performance? *(CO1)*

(OR)

b) Evaluate the suitability of the orthotropic plate method for analyzing bridge decks spanning in two directions under heavy vehicular loads? *(CO3)*

Q4 **Answer any one.** **(1X10=10 Marks)**

(a) Demonstrate your considerations to calculate the maximum Shear Force for a Beam span of 8m for IRC Class 70R Wheeled Loads.? *(CO1)*

(OR)

(b) A simply supported RC slab bridge deck spanning 6 m is subjected to a IRC 70R tracked load. Calculate the maximum possible bending moment for the beam? *(CO3)*

Q5 **Answer any one.** **(1X10=10 Marks)**

(a) A two-lane bridge deck (8 m wide, span 20 m) is subjected to a concentrated load of 150 kN at midspan. The wheel impression dimensions are 230X180mm, determine the approximate punching shear and longitudinal bending moment at critical sections? *(CO1)*

(OR)

(b) A two-way slab bridge deck (span 8 m × 6 m) is subjected to a concentrated load of 120 kN at the center. Using dispersion along the span, estimate the transverse bending moment at midspan? *(CO3)*